TOR AMPLITUDE ADDS OVER THE DERIVED TENSOR PRODUCT OF COMPLEXES OVER TWO RINGS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- Tor amplitude of a complex, with the reduction that the complex may be assumed bounded with flat terms in the given interval. the author cites Lemma lemma-tor-amplitude at proof lines 17079-17081. Supplies the normal form of the first complex used in the rest of the proof.
- Associativity of the derived tensor product in the form identifying a triple tensor product with the total complex of a double complex. the author cites Homology, Remark homology-remark-triple-complex at proof lines 17086-17089. Produces the double complex to which the spectral sequence is applied.
- Spectral sequence of a double complex and the resulting bound on the cohomology of the total complex. the author cites Homology, Lemma homology-lemma-ss-double-complex at proof lines 17092-17095. Gives the degree bound that is the statement.

Source reference. Original lemma 0B66: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14059–14059. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 0654: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 7888–7888. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 08BI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10568–10568. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0130: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1418–1418. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $A \to B$ be a ring map. Let $K^\bullet$ and $L^\bullet$ be complexes of B-modules. Let $a, b, c, d \in \mathbf{Z}$. If*

(1) *$K^\bullet$ as a complex of B-modules has tor amplitude in $[a, b]$,*
(2) *$L^\bullet$ as a complex of A-modules has tor amplitude in $[c, d]$,*

then $K^\bullet \otimes^{\mathbf{L}}_B L^\bullet$ as a complex of A-modules has tor amplitude in $[a + c, b + d]$.

Proof. We may assume that $K^\bullet$ is a complex of flat B-modules with $K^i = 0$ for $i \notin [a, b]$, see Lemma 0654. Let M be an A-module. Choose a free resolution $F^\bullet \to M$. Then

$$(K^\bullet \otimes^{\mathbf{L}}_B L^\bullet) \otimes^{\mathbf{L}}_A M = \mathrm{Tot}(\mathrm{Tot}(K^\bullet \otimes_B L^\bullet) \otimes_A F^\bullet) = \mathrm{Tot}(K^\bullet \otimes_B \mathrm{Tot}(L^\bullet \otimes_A F^\bullet))$$

see Homology, Remark 08BI for the second equality. By assumption (2) the complex $\mathrm{Tot}(L^\bullet \otimes_A F^\bullet)$ has nonzero cohomology only in degrees $[c, d]$. Hence the spectral sequence of Homology, Lemma 0130 for the double complex $K^\bullet \otimes_B \mathrm{Tot}(L^\bullet \otimes_A F^\bullet)$ proves that $(K^\bullet \otimes^{\mathbf{L}}_B L^\bullet) \otimes^{\mathbf{L}}_A M$ has nonzero cohomology only in degrees $[a + c, b + d]$. □